Hardware Implementation of a Radix-2 FFT Accelerator in Verilog

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***Abstract*—This paper presents the design and verification of a 64-point radix-2 decimation-in-time (DIT) Fast Fourier Transform (FFT) accelerator. The architecture employs a single-cycle butterfly unit, dual-port SRAM for in-place computation, and a hardwired control unit implementing a hierarchical state machine. The design was modeled in Verilog HDL and verified through spectral analysis using impulse and sinusoidal inputs, achieving correct frequency isolation and theoretical peak magnitude accuracy.**

I. Introduction

The Discrete Fourier Transform (DFT) is a fundamental operation in digital signal processing, widely used in applications such as spectral analysis, filtering, and communication systems. The DFT of a discrete sequence $x[n]$ of length N is defined as

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad W_N = e^{-j\frac{2\pi}{N}} \tag{1}$$

Direct computation requires $O(N^2)$ operations, which is impractical for real-time hardware implementations at moderate values of N. The Fast Fourier Transform (FFT), introduced by Cooley and Tukey, reduces this complexity to $O(N \log_2 N)$.

This work implements a radix-2 decimation-in-time FFT due to its structural regularity, hardware simplicity, and compatibility with binary memory addressing.

II. Architecture Design

A. Datapath

The datapath architecture implements a radix-2 butterfly capable of performing complex addition, subtraction, and twiddle-factor multiplication. Dual-port SRAM enables simultaneous read and write operations, allowing in-place computation without auxiliary buffers.

B. Control Unit

The control unit generates memory addresses and controls the FFT traversal order using a hardwired finite state machine. Three nested counters (stage, group, and pair) implement the radix-2 DIT flow.

The memory addresses for each butterfly operation are computed as

$$\text{Addr}_A = (\text{Group} \times 2 \times \text{Span}) + \text{Pair} \tag{2}$$

$$\text{Addr}_B = \text{Addr}_A + \text{Span} \tag{3}$$

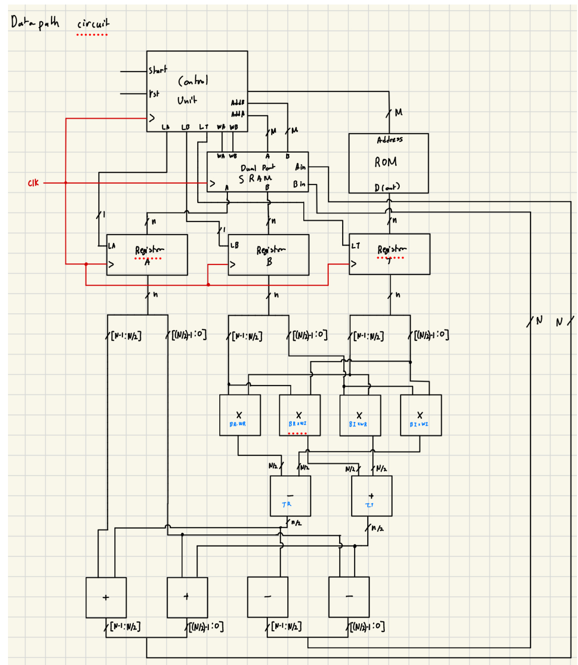

Fig. 1. RTL datapath architecture implementing an in-place radix-2 FFT with a shared complex multiplier and feedback path.

III. Implementation Details

The design was implemented entirely in Verilog HDL. To prevent read-after-write hazards during single-cycle butterfly execution, latch-enable signals were used to freeze memory outputs prior to write-back.

Listing 1. Control Unit Address Generation Logic

```
wire [M-1:0] group_offset;
assign group_offset = group_cnt * (
    span << 1);
assign addr_a = group_offset +
    pair_cnt;
assign addr_b = addr_a + span;
```

Bit-reversed input loading was performed prior to computation, eliminating the need for output reordering.

IV. Simulation and Verification

Verification was performed using functional simulation and waveform inspection. Two test cases were evaluated:

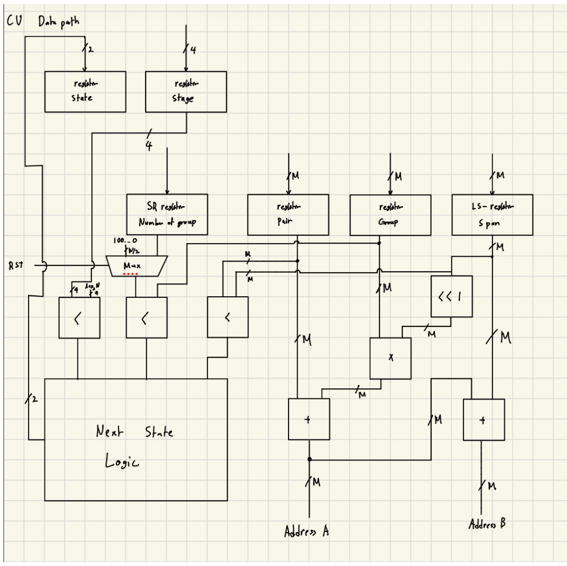

Fig. 2. Control unit schematic showing hierarchical counters and combinational address generation logic.

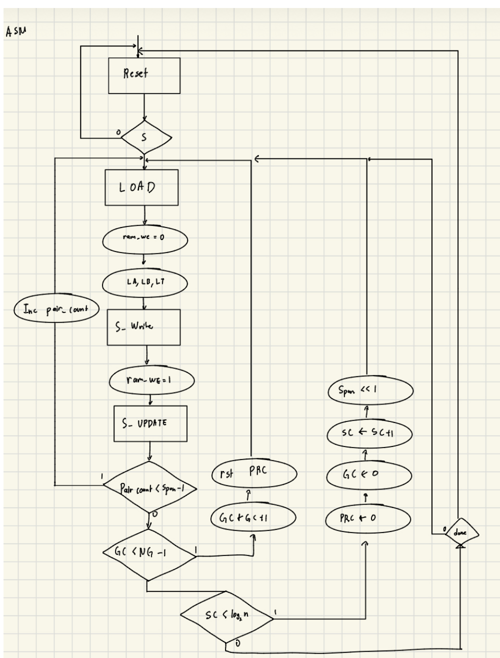

Fig. 3. Algorithmic state machine illustrating nested stage, group, and pair traversal for radix-2 DIT FFT execution.

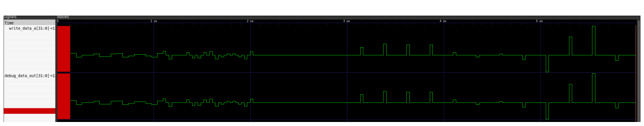

Fig. 4. Simulation results. Top: 64-point discrete cosine input signal. Bottom: FFT magnitude output showing spectral peaks at bins 4 and 60.

1) Unit impulse input, producing a constant-magnitude frequency spectrum.
2) A cosine wave with frequency index $k = 4$, producing spectral peaks at bins $k = 4$ and $k = 60$.

The measured peak magnitude matches theoretical expectations:

$$\text{Peak} = \frac{A \times N}{2} = \frac{1000 \times 64}{2} = 32000 \tag{4}$$

V. Conclusion

A 64-point radix-2 FFT accelerator was successfully designed and verified using Verilog HDL. The architecture achieves in-place computation using minimal hardware resources while maintaining correct spectral behavior. Future work includes FPGA synthesis, fixed-point optimization, and throughput benchmarking.